Kubernetes Components

Simple Definition

Kubernetes components are the different parts of Kubernetes that work together to manage containerized applications.

Think of Kubernetes as a **school**.

- 🏫 School = Kubernetes Cluster
- 👨‍💼 Principal = Control Plane
- 👨‍🏫 Teachers = Worker Nodes
- 👨‍🎓 Students = Pods

Every person has a different job.

Similarly, every Kubernetes component has a different responsibility.

Kubernetes Components

There are **2 main components**:

Kubernetes Cluster

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├── 1. Control Plane (Master)

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└── 2. Worker Node(s)

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1. Control Plane (Master)

Simple Definition

The **Control Plane** is the **brain** of Kubernetes.

It **does not run applications**.

It only **manages** everything.

Responsibilities

- Controls the entire cluster.
- Decides where Pods should run.
- Monitors Worker Nodes.
- Restarts failed Pods.
- Receives user requests.

Easy Example

🏫 School Principal

The principal:

- Gives instructions.
- Checks teachers.
- Solves problems.

The principal doesn't teach students.

Similarly,

Control Plane **manages** but **doesn't run applications**.

Components Inside the Control Plane

There are **4 important components**.

A. API Server
Simple Definition
API Server is the **main entry point** of Kubernetes.
Every command first goes here.
Responsibilities

- Receives user commands.
- Communicates with all Kubernetes components.

Example
You type:
kubectl create deployment
↓
API Server receives the request.
Easy Memory

🚪 **API Server = Front Door of Kubernetes**

B. Scheduler

Simple Definition

Scheduler decides **which Worker Node will run a Pod**.

Responsibilities

- Finds a Worker Node with enough resources.
- Assigns Pods to that node.

Example

Worker Node 1 → Busy ❌

Worker Node 2 → Free ✅

Scheduler chooses Worker Node 2.

Easy Memory

📍 **Scheduler = Pod Placement Manager**

C. etcd

Simple Definition

etcd is the **database** of Kubernetes.
It stores all cluster information.

Stores

- Nodes
- Pods
- Services
- Configurations
- Cluster state

Easy Memory

💾 **etcd = Kubernetes Database**

D. Controller Manager

Simple Definition

Controller Manager keeps checking that everything is running correctly.

Responsibilities

- Watches the cluster.
- Detects failed Pods.
- Creates new Pods if needed.

Example

You want:
3 Pods
One Pod crashes.
↓
Controller Manager creates another Pod.

Easy Memory
👮 **Controller Manager = Health Checker**

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2. Worker Node

Simple Definition

Worker Node is the machine where applications actually run.

Responsibilities

- Runs Pods.
- Runs Containers.
- Executes applications.

Easy Example

👨‍🏫 Teacher

Teacher actually teaches students.

Similarly,

Worker Node actually runs applications.

Components Inside Worker Node

There are **3 important components**.

A. Kubelet
Simple Definition
Kubelet is an **agent** running on every Worker Node.
Responsibilities

- Receives instructions from the Control Plane.
- Starts Pods.
- Monitors Pods.
- Reports status back.

Easy Memory
📞 **Kubelet = Messenger between Worker Node and Control Plane**

B. Container Runtime **Simple Definition** Container Runtime runs the containers. Examples: • containerd • CRI-O (Docker was used as a runtime in older Kubernetes versions.) **Easy Memory** 📦 **Container Runtime = Container Runner**
C. Kube Proxy **Simple Definition** Kube Proxy manages networking inside the cluster. **Responsibilities** • Sends traffic to the correct Pod. • Enables communication between services and Pods. **Easy Memory** 🚦 **Kube Proxy = Traffic Police**

Pod

Simple Definition

A **Pod** is the smallest deployable unit in Kubernetes.

Usually,

One Pod contains one container.

Pod

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└── Container

|

└── Application

Easy Memory

📦 **Pod = Box that holds Container(s)**

Container

Simple Definition

A Container contains the application and everything it needs to run.

Created by:

- Docker
- containerd
- CRI-O

Easy Memory

📦 **Container = Application Package**

Complete Flow

User

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API Server

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Scheduler

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Worker Node

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Kubelet

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Pod

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Container Runtime

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Container

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Application

Quick Revision Table

Component	**Easy Meaning**
Cluster	Group of Nodes
Control Plane	Brain of Kubernetes
API Server	Front Door (Receives Requests)
Scheduler	Chooses Worker Node
etcd	Kubernetes Database
Controller Manager	Health Checker (Self-Healing)
Worker Node	Runs Applications
Kubelet	Worker Node Agent
Container Runtime	Runs Containers
Kube Proxy	Manages Networking
Pod	Smallest Deployable Unit

Component	Easy Meaning
Container	Holds the Application

Easy Trick to Remember ⭐

Control Plane (Brain)

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├── API Server → Front Door 🚪

├── Scheduler → Pod Placement 📍

├── etcd → Database 💾

└── Controller Manager → Health Checker 👮

Worker Node

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├── Kubelet → Agent 📞

├── Container Runtime → Runs Containers 📦

└── Kube Proxy → Traffic Police 🚦

Pod → Holds Container 📦

Container → Runs Application 💻

One-Line Memory Trick

- 🧠 **Control Plane = Manages the cluster**
- 🚪 **API Server = Entry point**
- 📍 **Scheduler = Chooses where Pods run**
- 💾 **etcd = Stores cluster data**

- 👮 **Controller Manager = Maintains the desired state**
- 💻 **Worker Node = Runs the application**
- 📞 **Kubelet = Talks to the Control Plane**
- 📦 **Container Runtime = Runs containers**
- 🚦 **Kube Proxy = Handles network traffic**
- 📦 **Pod = Smallest unit Kubernetes deploys**